सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

Qualification Pack

Plastic Mold Design and Manufacturing Engineer

QP Code: CUTM/CSC/Q0203

Version: 1.0

NSQF Level: 6

Centurion University of Technology and Management || Aluri Nagar, Paralakhemundi, R Sitapur Uppalada || email:partha@cutm.ac.in

Contents

CUTM/CSC/Q0203: Plastic Mold Design and Manufacturing Engineer 3
Brief Job Description 3
Applicable National Occupational Standards (NOS) 3
Compulsory NOS 3
Qualification Pack (QP) Parameters 3
CUTM/CSC/N0209: Interpretation of designs and drawings and process planning. 5
CUTM/CSC/N0210: CAM Programming and setting equipment, machining and maintenance 9
CUTM/CSC/N0211: Assemble, try out of mold, and inspect plastic parts 14
CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place 19
DGT/VSQ/N0102: Employability Skills (60 Hours) 23
Assessment Guidelines and Weightage 30
Assessment Guidelines 30
Assessment Weightage 32
Acronyms 33
Glossary 34

Qualification Pack

CUTM/CSC/Q0203: Plastic Mold Design and Manufacturing Engineer

Brief Job Description

Plastic Mold Design and Manufacturing Engineer is responsible for product modeling, designing mold, generating CAM design for manufacturing of various mold parts, machine mold parts, assemble and pilot testing. Within this sector, plastic die technicians, mold makers, carry out a range of different duties along with maintenance and repair of equipment.

Personal Attributes

Personal attributes for a Plastic Mold Design and Manufacturing Engineer include strong analytical skills for interpreting designs and drawings, attention to detail in process planning, proficiency in CAM programming, and expertise in machine setup and maintenance. They should have a hands-on approach to mold assembly, testing, and inspection, ensuring high-quality plastic parts are produced and meet design specifications.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. CUTM/CSC/N0209: Interpretation of designs and drawings and process planning.
2. CUTM/CSC/N0210: CAM Programming and setting equipment, machining and maintenance
3. CUTM/CSC/N0211: Assemble, try out of mold, and inspect plastic parts
4. CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place
5. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Plastic Mold
Occupation	Machining
Country	India
NSQF Level	6
Credits	20

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	Completed 3 year UG degree (B.E. or B. Tech degree in Mechanical or Production or Mechatronics) with 1 Year of experience relevant experience OR Completed 2nd year diploma after 12th (Mechanical or Production) with 3 Years of experience relevant experience OR Completed 3 year diploma after 10th (Mechanical or Production) with 4.5 years of experience relevant experience OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	30/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-06-CG-01977-2024-V1-CUTM
NQR Version	1.0

CUTM/CSC/N0209: Interpretation of designs and drawings and process planning.

Description

This NOS covers the interpretation of mold designs and engineering drawings and the development of a structured process plan for mold manufacturing. It involves analyzing design specifications, material selection, machining sequences, and quality control requirements to ensure accurate mold fabrication while optimizing efficiency and production feasibility.

Scope

The scope covers the following :

- The scope includes analyzing and interpreting mold drawings, understanding GD&T principles, planning machining and assembly processes, and ensuring manufacturability. It also covers tool selection, work instructions, BOM preparation, and quality compliance while coordinating with designers, engineers, and production teams to execute process plans effectively.

Elements and Performance Criteria

Interpretation of designs and drawings

To be competent, the user/individual on the job must be able to:

PC1. interpret technical drawings and specifications of product.

PC2. identify critical features of product, connections, and geometric requirements.

PC3. analyze manufacturability with available resources.

PC4. identify and prepare for potential assembly issues if any

PC5. identify and prepare for any maintenance issues that may arise during production.

PC6. specify stock to be kept for different operations.

PC7. plan production of parts according to specifications.

Process Planning

To be competent, the user/individual on the job must be able to:

PC8. importance of planning to improve efficiency.

PC9. procedures to manufacture molds with available resources.

PC10. machining operations and their sequences.

PC11. method of clamping work pieces.

PC12. choices of cutting tools and cutting parameters.

PC13. machine and work piece setting.

PC14. measuring tools and equipment.

PC15. bench work and assembly techniques.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. engineering drawings and mold design specifications.
KU2. CAD software (AutoCAD, SolidWorks, CATIA) for interpreting and modifying mold designs.
KU3. orthographic projections, sectional views, and dimensioning used in mold design.
KU4. plastic molding processes such as injection molding, compression molding, and blow molding.
KU5. plastic molding processes such as injection molding, compression molding, and blow molding.
KU6. plastic molding processes such as injection molding, compression molding, and blow molding.
KU7. BOM (Bill of Materials), process sheets, and work instructions for mold production.
KU8. BOM (Bill of Materials), process sheets, and work instructions for mold production.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. interpret engineering drawings and translate them into manufacturing processes.
GS2. coordinate with design teams, machinists, and production engineers.
GS3. Use decision-making abilities in selecting the appropriate machining and assembly processes.
GS4. Plan and organize to ensure smooth workflow and resource optimization.
GS5. Adapt, learn and applying new mold design and planning software.
GS6. Documentation skills for maintaining design modifications, process sheets, and inspection reports.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Interpretation of designs and drawings	**17**	**22**	**-**	**11**
PC1. interpret technical drawings and specifications of product.	2	3	-	2
PC2. identify critical features of product, connections, and geometric requirements.	2	3	-	2
PC3. analyze manufacturability with available resources.	3	4	-	-
PC4. identify and prepare for potential assembly issues if any	2	3	-	3
PC5. identify and prepare for any maintenance issues that may arise during production.	3	3	-	1
PC6. specify stock to be kept for different operations.	2	3	-	2
PC7. plan production of parts according to specifications.	3	3	-	1
Process Planning	**13**	**18**	**-**	**19**
PC8. importance of planning to improve efficiency.	1	2	-	4
PC9. procedures to manufacture molds with available resources.	1	1	-	4
PC10. machining operations and their sequences.	1	1	-	5
PC11. method of clamping work pieces.	1	2	-	3
PC12. choices of cutting tools and cutting parameters.	1	3	-	2
PC13. machine and work piece setting.	3	3	-	1
PC14. measuring tools and equipment.	2	3	-	-
PC15. bench work and assembly techniques.	3	3	-	-
NOS Total	**30**	**40**	**-**	**30**

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0209
NOS Name	Interpretation of designs and drawings and process planning.
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

CUTM/CSC/N0210: CAM Programming and setting equipment, machining and maintenance

Description

This NOS covers CAM programming, CNC machine setup, machining operations, and maintenance for plastic mold manufacturing. It includes generating G-code and M-code, selecting cutting tools, setting up workpieces, optimizing toolpaths, and performing machine maintenance to ensure precision machining and high-quality mold production.

Scope

The scope covers the following :

- The scope includes developing CAM programs, setting up CNC machines, selecting machining parameters, and executing cutting operations. It also covers machine calibration, tool offset adjustments, maintenance procedures, troubleshooting machining issues, and ensuring compliance with quality standards in mold manufacturing.

Elements and Performance Criteria

Programming and setting up equipment to fabricate mold

To be competent, the user/individual on the job must be able to:

PC1. select best sequence for machining each specific work piece.

PC2. program manually and in CAM software.

PC3. transfer programs to machines.

PC4. set work pieces and tools.

Machining of mold parts

To be competent, the user/individual on the job must be able to:

PC5. apply principles and processes of Computer Aided Manufacturing (CAM).

PC6. set up and use machine centre input data into CNC machine controllers (tool offset, work offset, etc.)

PC7. create machining programs in software, and transfer to machine controllers.

PC8. test finished products and assessed them for accuracy in accordance with specified drawings.

PC9. machine each part of dies, taking account of each plastic product’s requirements.

PC10. measure pieces of work accurately.

PC11. set offsets according to measured size.

PC12. achieve required geometry and finish.

PC13. fabricate all parts to commercial standards using machine centres, pull off grinders, drilling machines, bench grinders.

Maintenance and repair of machining equipment

To be competent, the user/individual on the job must be able to:

PC14. do preventative maintenance to avoid issues arising.

PC15. make repairs when problem arise.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. CAM (Computer-Aided Manufacturing) programming, including toolpath generation and optimization.
KU2. CNC machine operations, including milling, turning, and EDM (Electrical Discharge Machining).
KU3. G-codes and M-codes used for CNC programming and machine control.
KU4. cutting tool selection, feeds, speeds, and depth of cut for different materials.
KU5. work holding devices, fixtures, and clamping techniques for mold machining.
KU6. preventive maintenance schedules for CNC machines to ensure high performance.
KU7. machine calibration, tool offset adjustments, and troubleshooting CNC errors.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. interpret technical drawings and convert them into CAM programs.
GS2. coordinate with machine operators, engineers, and quality inspectors.
GS3. use problem-solving skills for debugging CAM programs and troubleshooting machining errors.
GS4. take decision on choosing the appropriate tools, speeds, and machining parameters.
GS5. Plan and organize machine setup, tool changes, and workflow optimization.
GS6. evaluate machining performance and improve efficiency.
GS7. work in a team with full teamwork and leadership skills in coordinating CNC machining and maintenance activities.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Programming and setting up equipment to fabricate mold	**12**	**14**	**-**	**-**
PC1. select best sequence for machining each specific work piece.	3	4	-	-
PC2. program manually and in CAM software.	3	4	-	-
PC3. transfer programs to machines.	3	3	-	-
PC4. set work pieces and tools.	3	3	-	-
Machining of mold parts	**24**	**38**	**-**	**-**
PC5. apply principles and processes of Computer Aided Manufacturing (CAM).	3	4	-	-
PC6. set up and use machine centre input data into CNC machine controllers (tool offset, work offset, etc.)	3	4	-	-
PC7. create machining programs in software, and transfer to machine controllers.	3	4	-	-
PC8. test finished products and assessed them for accuracy in accordance with specified drawings.	3	4	-	-
PC9. machine each part of dies, taking account of each plastic product’s requirements.	3	4	-	-
PC10. measure pieces of work accurately.	3	4	-	-
PC11. set offsets according to measured size.	2	4	-	-
PC12. achieve required geometry and finish.	2	4	-	-
PC13. fabricate all parts to commercial standards using machine centres, pull off grinders, drilling machines, bench grinders.	2	6	-	-
Maintenance and repair of machining equipment	**4**	**8**	**-**	**-**
PC14. do preventative maintenance to avoid issues arising.	2	4	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. make repairs when problem arise.	2	4	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0210
NOS Name	CAM Programming and setting equipment, machining and maintenance
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	7
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

CUTM/CSC/N0211: Assemble, try out of mold, and inspect plastic parts

Description

This NOS focuses on assembling molds, conducting mold trials, and inspecting plastic parts to ensure functionality and quality. It involves mold fitting, machine setup, defect analysis, and quality inspection while coordinating with design and production teams to achieve defect-free molded parts.

Scope

The scope covers the following :

- The scope includes assembling mold components, conducting mold trials, troubleshooting molding defects, and inspecting plastic parts for quality assurance. It also covers mold adjustments, machine parameter optimization, functional testing of plastic parts, and documentation to ensure the production of high-precision molded components.

Elements and Performance Criteria

Assembled mold parts

To be competent, the user/individual on the job must be able to:

PC1. interpret the mold assembly drawing.
PC2. create a sequence of action for mold assembly.
PC3. check the quality and quantity requirement of mold parts as per drawing.
PC4. ensure use of range of hand and power tools for assembly.
PC5. ensure proper techniques, tools, and methods for production line tools.
PC6. ascertain components are polished using polishing tools.
PC7. carry out drilling components as per design requirement.
PC8. implement principles of pin cutting and surface contact.
PC9. assemble components in preparation for testing.

Try out of assembled mold

To be competent, the user/individual on the job must be able to:

PC10. ensure to change various parameters in tryout of assembled mold.
PC11. execute machine in semi-automatic mode.

Inspect plastic parts produced using assembled mold

To be competent, the user/individual on the job must be able to:

PC12. identify defects in plastic products.
PC13. propose solutions for identified defects.
PC14. implement proposed solutions, if required.
PC15. accurately measure dimensions of products.
PC16. ensure to achieve product dimensions as in drawings or models.
PC17. check condition of both interiors and exteriors of products.
PC18. modify molding parameters and develop plastic products.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. mold assembly procedures, including fitting cores, cavities, ejector systems, and cooling channels.
KU2. different types of molds, such as two-plate, three-plate, and hot runner systems.
KU3. injection molding machines and their setup for mold tryouts.
KU4. plastic material properties and their behavior during molding.
KU5. mold alignment, clamping techniques, and sealing methods to prevent defects.
KU6. troubleshooting common molding defects, including flash, sink marks, warpage, and short shots.
KU7. inspection techniques for plastic parts, including visual inspection, dimensional measurement, and functional testing.
KU8. safety protocols in mold handling, assembly, and machine operation.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. safety protocols in mold handling, assembly, and machine operation.
GS2. coordinate with design, production, and quality teams.
GS3. use problem-solving skills for identifying and correcting mold assembly or plastic part defects.
GS4. Plan and organize mold assembly, tryout sequences, and troubleshooting steps.
GS5. coordinate with machine operators, mold technicians, and quality inspectors.
GS6. implement new mold assembly and inspection techniques.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Assembled mold parts	**18**	**27**	**-**	**-**
PC1. interpret the mold assembly drawing.	2	3	-	-
PC2. create a sequence of action for mold assembly.	2	3	-	-
PC3. check the quality and quantity requirement of mold parts as per drawing.	2	3	-	-
PC4. ensure use of range of hand and power tools for assembly.	2	3	-	-
PC5. ensure proper techniques, tools, and methods for production line tools.	2	3	-	-
PC6. ascertain components are polished using polishing tools.	2	3	-	-
PC7. carry out drilling components as per design requirement.	2	3	-	-
PC8. implement principles of pin cutting and surface contact.	2	3	-	-
PC9. assemble components in preparation for testing.	2	3	-	-
Try out of assembled mold	**8**	**12**	**-**	**-**
PC10. ensure to change various parameters in tryout of assembled mold.	5	7	-	-
PC11. execute machine in semi-automatic mode.	3	5	-	-
Inspect plastic parts produced using assembled mold	**14**	**21**	**-**	**-**
PC12. identify defects in plastic products.	2	3	-	-
PC13. propose solutions for identified defects.	2	3	-	-
PC14. implement proposed solutions, if required.	2	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. accurately measure dimensions of products.	2	3	-	-
PC16. ensure to achieve product dimensions as in drawings or models.	2	3	-	-
PC17. check condition of both interiors and exteriors of products.	2	3	-	-
PC18. modify molding parameters and develop plastic products.	2	3	-	-
NOS Total	**40**	**60**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0211
NOS Name	Assemble, try out of mold, and inspect plastic parts
Sector	Capital Goods
Sub-Sector	
Occupation	Machining
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

CUTM/VSQ/N7805: Ensure and Implement Safety standards at Work Place

Description

This NOS focuses on developing the ability to operate CNC machining centers safely and efficiently while adhering to health and safety standards. It includes creating CAM programs, selecting tools and cutting parameters, maintaining machinery, communicating effectively with teams and clients, and continuously upgrading skills to ensure excellence in manufacturing processes.

Scope

The scope covers the following :

- The scope encompasses CNC machine operation, safety compliance, program development, tool selection, maintenance, and effective collaboration with team members and customers. It promotes high professional standards through ongoing skill enhancement, problem-solving, and customer engagement while maintaining health and safety protocols throughout the manufacturing lifecycle.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. adhere to health and safety regulations in the workplace
PC2. proactively promote best practice in health and safety in the working environment
PC3. work independently on CNC machining centres
PC4. create manual and CAM programs for various types of machining
PC5. select suitable cutting parameters
PC6. select and set the most appropriate tools for the planned work
PC7. maintain all tools to ensure that they are in the best condition
PC8. communicate and collaborate effectively with colleagues, team members, and other professionals
PC9. engage with customers effectively, always prioritizing their needs
PC10. explain complex technical details to non-specialists
PC11. proactively engage in continuous professional development to promote excellence in the work and maintain expertise in current industrial practice analyze the manufacturing feasibility
PC12. successfully apply mathematical principles to complex industrial scenarios
PC13. demonstrate high levels of critical thinking

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Workplace health and safety regulations, and how to adhere to them.
KU2. Best practices in promoting health and safety proactively in CNC operations.

KU3. Fundamentals of CNC machining processes and the safe operation of CNC machines.
KU4. CAM software programming principles and manual programming techniques.
KU5. Selection of appropriate cutting parameters and tools for various materials and tasks.
KU6. Maintenance procedures for tools and machines to ensure optimal performance and safety.
KU7. Communication protocols and collaborative practices within a technical team.
KU8. Techniques for engaging with customers and understanding their technical needs.
KU9. Simplifying and explaining technical information to non-specialists.
KU10. Importance of continuous professional development and industrial best practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Independently operate CNC machines with precision and safety.
GS2. Develop CAM programs suited for diverse machining requirements.
GS3. Select and calibrate tools and cutting parameters for optimal machining.
GS4. Maintain tools and machines in excellent working condition.
GS5. Collaborate effectively with teammates and interdisciplinary professionals.
GS6. Provide clear, simple explanations of technical content to clients or stakeholders.
GS7. Prioritize customer requirements and ensure satisfaction in service delivery.
GS8. Apply mathematical reasoning to solve complex machining problems.
GS9. Analyze the feasibility of manufacturing processes and adapt accordingly.
GS10. Exhibit strong critical thinking and decision-making in industrial scenarios.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	**40**	**-**	**30**
PC1. adhere to health and safety regulations in the workplace	2	3	-	2
PC2. proactively promote best practice in health and safety in the working environment	2	3	-	2
PC3. work independently on CNC machining centres	2	3	-	2
PC4. create manual and CAM programs for various types of machining	2	3	-	2
PC5. select suitable cutting parameters	2	3	-	2
PC6. select and set the most appropriate tools for the planned work	2	3	-	2
PC7. maintain all tools to ensure that they are in the best condition	2	3	-	2
PC8. communicate and collaborate effectively with colleagues, team members, and other professionals	2	3	-	2
PC9. engage with customers effectively, always prioritizing their needs	2	2	-	2
PC10. explain complex technical details to non-specialists	2	2	-	3
PC11. proactively engage in continuous professional development to promote excellence in the work and maintain expertise in current industrial practice analyze the manufacturing feasibility	3	4	-	3
PC12. successfully apply mathematical principles to complex industrial scenarios	3	4	-	3
PC13. demonstrate high levels of critical thinking	4	4	-	3
NOS Total	**30**	**40**	**-**	**30**

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7805
NOS Name	Ensure and Implement Safety standards at Work Place
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all

GS4. how to work in a virtual mode

GS5. perform calculations efficiently

GS6. solve problems effectively

GS7. pay attention to details

GS8. manage time efficiently

GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

• Batches assigned to the assessment agencies for conducting the assessment on SIDH or email

• Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH

• Assessment agency deploys the ToA certified Assessor for executing the assessment

• AB monitors the assessment process & records

2. Testing Environment:

• Check the Assessment location, date and time

• If the batch size is more than 30, then there should be 2 Assessors.

• Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

• Question bank is created by the Subject Matter Experts (SME) are verified by the AB

• Questions are mapped to the specified assessment criteria.

• Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

• Time-stamped & geotagged reporting of the assessor from assessment location.

• Centre photographs with signboards and scheme specific branding

• Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

• Surprise visit to the assessment location

6. Minimum Percentage Required

• 70% overall to pass.

• Minimum 70% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(**Please note**: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/CSC/N0209.Interpretation of designs and drawings and process planning.	30	40	-	30	100	25
CUTM/CSC/N0210.CAM Programming and setting equipment, machining and maintenance	40	60	-	-	100	25
CUTM/CSC/N0211.Assemble, try out of mold, and inspect plastic parts	40	60	-	-	100	25
CUTM/VSQ/N7805.Ensure and Implement Safety standards at Work Place	30	40	-	30	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	**160**	**230**	**-**	**60**	**450**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.